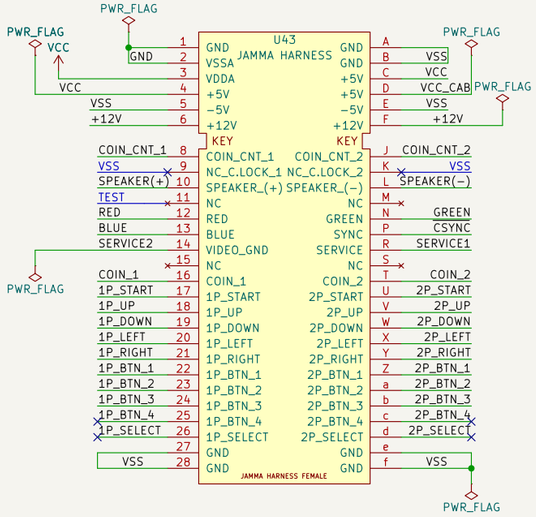
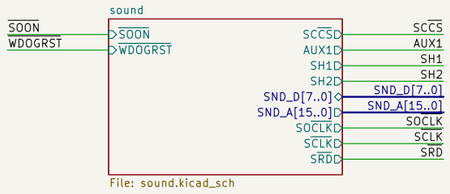
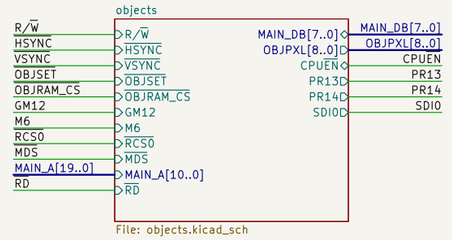
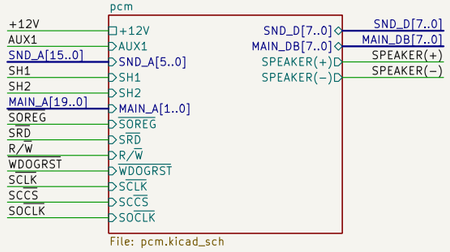
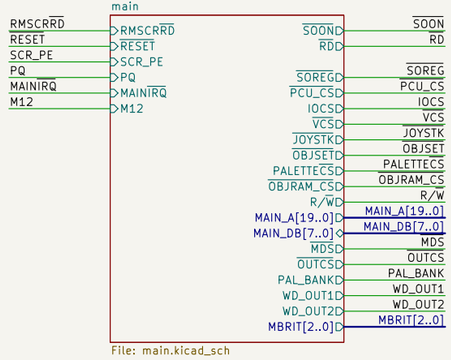
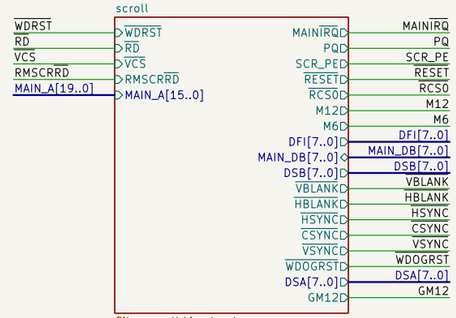


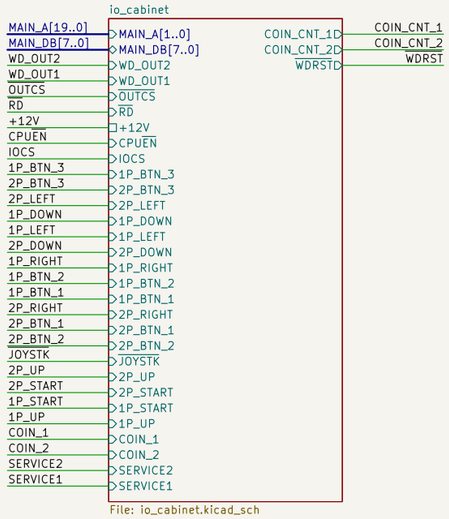


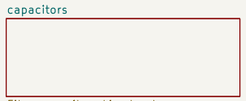


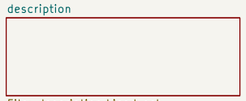


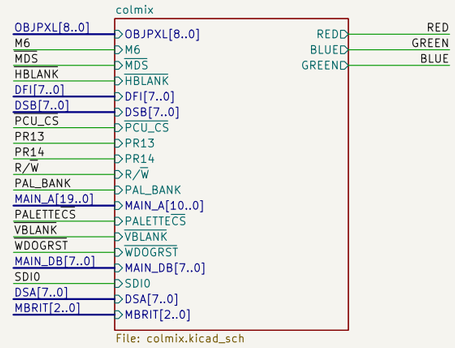












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Sheet: /
File: parodius.kicad_sch

Title: Parodius Da!

Size: A3	Date: 2023-05-09	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 1/11

MAIN_D[7..0]

TEST HEADER
FOOTPRINT MAIN CPU

Conn_02x25_Counter_Clockwise
FP2

Conn_02x25_Counter_Clockwise
FP1

4728 IAM
RA14
4.7kΩx8

F19B 74LS74
F11C 74LS08
F19A 74LS74
74LS08 F11B
H21E 74LS04
H22A 74LS74
H21F 74LS04

H7 74LS245
buffers
E16C 74LS00

F10 74LS367
3-StateHexBuffers

H12
KONAMI
053248
CPU MAIN

MAIN_A[19..0]

F5 27C010
H5 27C010

NOT CONNECTED

PAL2
PROG_A13= PAGED ? A15 : A13
PROG_A14= PAGED ? A16 : A14
PROG_A15= PAGED ? A17 : A15
PROG_A16= PAGED ? A18 : 0
PROGCS1 = A19 & PAGED
PROGCS2 = UNPAGED & (A19 | PAGED)

PAL2 PAL16L8

Floating (no driver)
verified on the PCB
while running the game
Constant 1.8V observed

MBRIT[2..0]

flip flops 74LS273

F12A 74LS32

PAL3 PAL16L8
PAL3
o12 = i7 = 2000-2FFF when BANK1
2800-5FFF
6000-7FFF when BANKR
o13 = i6 = 3C00-3FFF
PAGED= 6000-7FFF when BANKR
8000-9FFF
PALETTE_CS= 0000-07FF when WRK
OBJRAM_CS= 2000-27FF when BANK1
WORKCS= 0000-07FF when WRK
0800-1FFF
UNPAGED= A000-FFFF (ROM PPAL)

PAL1 PAL16L8
PAL1
JOYSTK 3F8C-3F8F
IO_CS 3F90-3F9F
MISC 3FC0-3FCF
PCU_CS 3FB0-3FBF
OBJSET 3FA0-3FAF
VCS= AS&i7&(A8&A7&A9&AB+i6+A7)

Connected to pin 7
of alternate
YM2151 footprint.

74LS139 decoder

E17 MB8464A-10W-15W
CMOS Static RAM
8K x 8 BITS

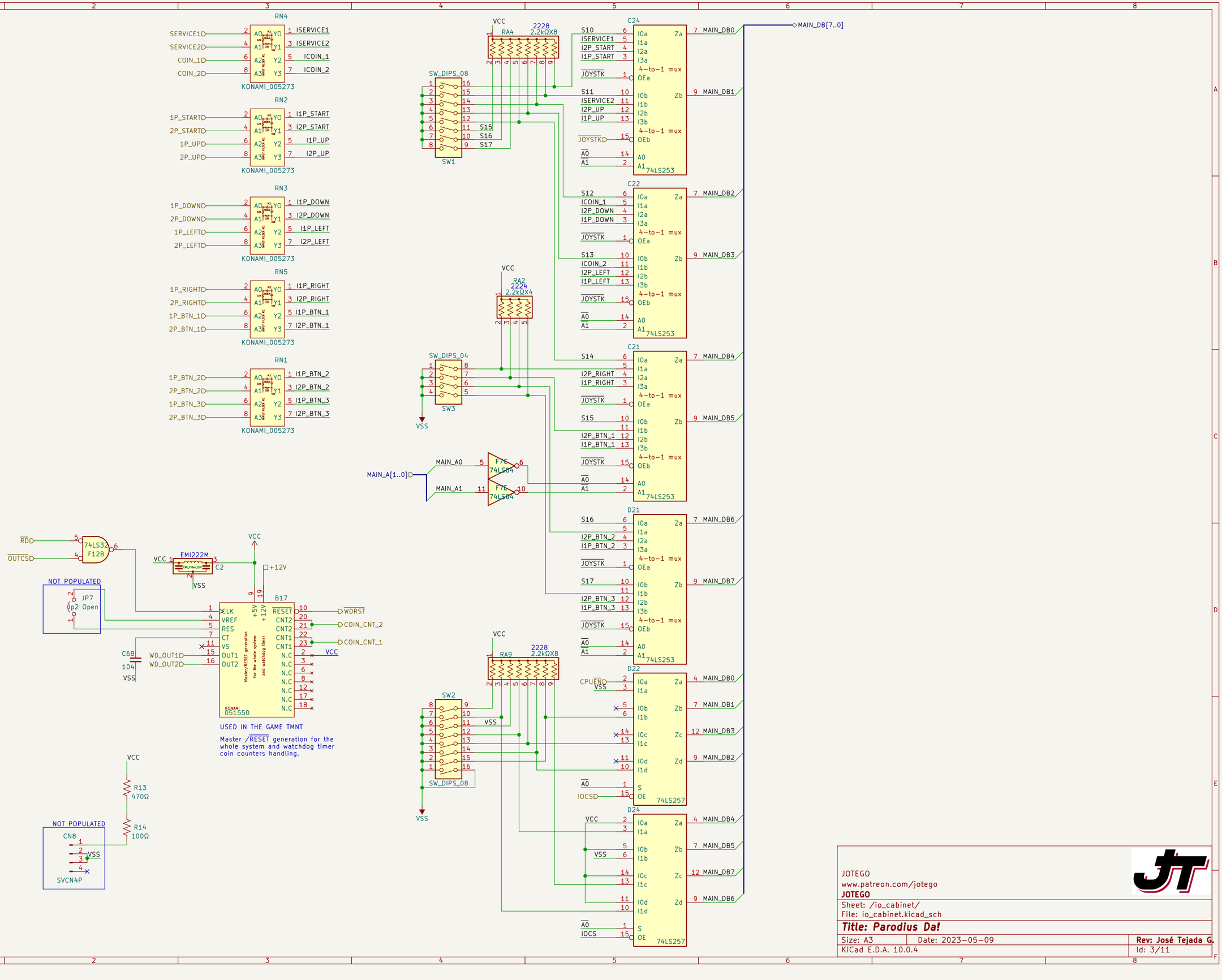
hex flip flop 74LS174

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Sheet: /main/
File: main.kicad_sch

Title: Parodius Da!

Size: A3	Date: 2023-05-09	Rev: José Tejada G.
KiCad E.D.A. 10.0.4		Id: 2/11


RN4
SERVICE1
SERVICE2
COIN_1
COIN_2
ISERVICE1
ISERVICE2
ICOIN_1
ICOIN_2
KONAMI_005273
RN2
1P_START
2P_START
1P_UP
2P_UP
I1P_START
I2P_START
I1P_UP
I2P_UP
KONAMI_005273
RN3
1P_DOWN
2P_DOWN
1P_LEFT
2P_LEFT
I1P_DOWN
I2P_DOWN
I1P_LEFT
I2P_LEFT
KONAMI_005273
RN5
1P_RIGHT
2P_RIGHT
1P_BTN_1
2P_BTN_1
I1P_RIGHT
I2P_RIGHT
I1P_BTN_1
I2P_BTN_1
KONAMI_005273
RN1
1P_BTN_2
2P_BTN_2
1P_BTN_3
2P_BTN_3
I1P_BTN_2
I2P_BTN_2
I1P_BTN_3
I2P_BTN_3
KONAMI_005273
VCC
RA4
2228
2.2kΩX8
SW_DIPS_08
SW1
S15
S16
S17
C24
C22
C21
D21
D22
D24
74LS253
74LS257
VCC
RA2
2224
2.2kΩX4
SW_DIPS_04
SW3
VSS
MAIN_A[1..0]
MAIN_A0
MAIN_A1
74LS04
MAIN_DB[7..0]
MAIN_DB0
MAIN_DB1
MAIN_DB2
MAIN_DB3
MAIN_DB4
MAIN_DB5
MAIN_DB6
MAIN_DB7
RA9
SW2
SW_DIPS_08
RD
OUTC5
74LS32
F12B
EMI222M
C2
+12V
NOT POPULATED
JP7
Jp2 Open
B17
CLK
VREF
RES
CT
VS
OUT1
OUT2
RESET
CNT2
CNT2
CNT1
CNT1
+5V
+12V
N.C
Master/RESET generation
for the whole system
and watchdog timer
KONAMI
051550
WDRST
COIN_CNT_2
COIN_CNT_1
C68
104
WD_OUT1
WD_OUT2
USED IN THE GAME TMNT
Master /RESET generation for the
whole system and watchdog timer
coin counters handling.
R13
470Ω
R14
100Ω
CN8
SVCN4P
CPUEN
IOCS
JOTEGO
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JOTEGO
Sheet: /io_cabinet/
File: io_cabinet.kicad_sch
Title: Parodius Da!
Size: A3
Date: 2023-05-09
Rev: José Tejada G.
KiCad E.D.A. 10.0.4
Id: 3/11

NOT POPULATED

RA8 4kX8 RA13 4kX8 RA12 4kX8 RA10 4kX8

GFXROMA[17..0]

CN7

K13 955005-06-07-08 KONAMI MASK 4M - 16BIT HN62414P CMOS ROM

K19 955005-06-07-08 KONAMI MASK 4M - 16BIT HN62414P CMOS ROM

GFXROMD[31..0]

DSA[7..0] SCRA

FIX DFI[7..0]

J15 KONAMI 051962 TILE DRAWING

LAYER A COLOR INDEX · GFX ROM DATA · LAYER A TRANSPARENCY · FIX LAYER TRANSPARENCY · FIX LAYER COLOR INDEX · CPU DATA TO READ GFX ROMs · LAYER B TRANSPARENCY · LAYER B COLOR INDEX · ATTRIBUTE CODE FROM 052109

MAIN_DB[7..0]

DSB[7..0] SCRB

C15C 74LS07 · R18 470Ω · R20 100Ω · CSYNC

JP6 Jp_2_Bridged

G22 XTAL_OSC 24MHz · CP1 104

74LS04 H21A · 74LS04 H21C

H22B 74LS74

WDRST · WDOGRST · WDRSTD

TEST Jp_2_Bridged JP5

MAIN_A[15..0]

J20 KONAMI 052109 TILE MAP GENERATOR

LAYER A SCROLL X · LAYER B SCROLL X · TILE ATTRIBUTES · GFX ROM ADDRESS · MAIN CLK · CPU ADDRESS · GFX ROM read · VRAM ADDRESS · VRAM DATA · CPU DATA

SCRAMA[12..0]

H24 J24 MB8464A-10W-15W CMOS Static RAM 8K x 8 BITS

VD[15..0]

MAIN_DB[7..0]

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Sheet: /scroll/
File: scroll.kicad_sch

Title: Parodius Da!

Size: A3 | Date: 2023-05-09 | Rev: José Tejada G.
KiCad E.D.A. 10.0.4 | Id: 4/11

NOT POPULATED

MAIN_A[10..0]

OBJROMA[18..0]

OBJROMD[31..0]

ROM's MASK OF THE SPRITES

U41
KONAMI
053245
OBJECT GENERATOR

U42
053244

K2
K8
955D05-06-07-08
KONAMI
MASK 4M - 16BIT
HN62414P
CMOS ROM

HM6116P-3
2kB RAM

74LS245
buffers

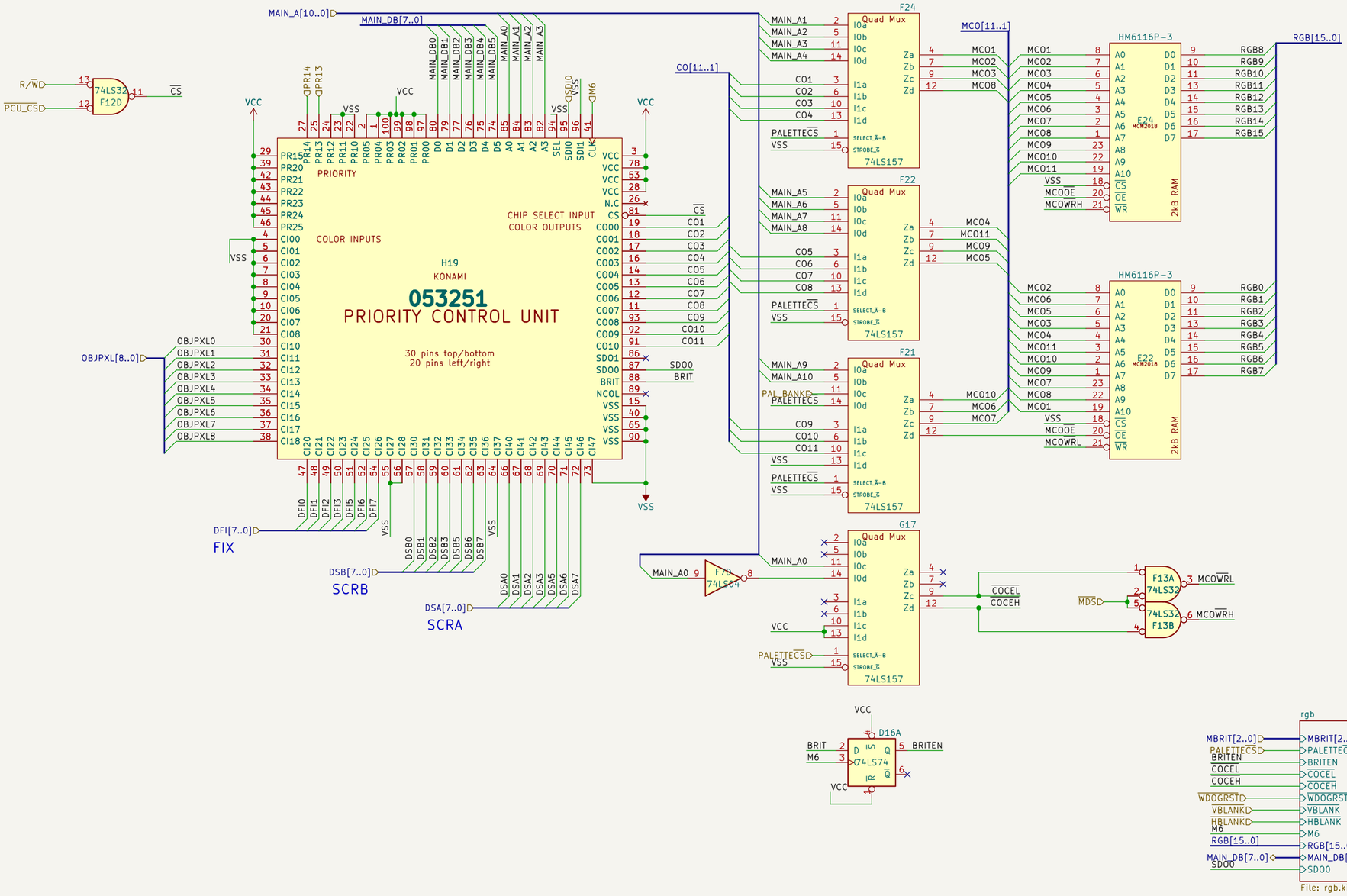
Conn_02x32_Row_Letter_First

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Sheet: /objects/
File: objects.kicad_sch

Title: Parodius Da!

Size: A3	Date: 2023-05-09	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 5/11



JOTEGO
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Sheet: /colmix/
File: colmix.kicad_sch

Title: Parodius Da!

Size: A3	Date: 2023-05-09	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 6/11

RGB[15..0]

MAIN_DB[7..0]

E21

MAIN_DB0 2 A0 B0 18 RGB0
MAIN_DB1 3 A1 B1 17 RGB1
MAIN_DB2 4 A2 B2 16 RGB2
MAIN_DB3 5 A3 B3 15 RGB3
MAIN_DB4 6 A4 B4 14 RGB4
MAIN_DB5 7 A5 B5 13 RGB5
MAIN_DB6 8 A6 B6 12 RGB6
MAIN_DB7 9 A7 B7 11 RGB7
buffers
PALETTECS 1 A->B
COCEL 19 CE
74LS245

RGB0 3 D0 Q0 2 G3
RGB1 4 D1 Q1 5 G4
RGB2 7 D2 Q2 6 B0
RGB3 8 D3 Q3 9 B1
RGB4 13 D4 D19 Q4 12 B2
RGB5 14 D5 Q5 15 B3
RGB6 17 D6 Q6 16 B4
RGB7 D7 Q7 19 SHADOW
flip flops
SDOOD 5 C14C 6 18
74LS04
M6 11 Cp
VCC 1 Mr
74LS273

E20

MAIN_DB0 2 A0 B0 18 RGB8
MAIN_DB1 3 A1 B1 17 RGB9
MAIN_DB2 4 A2 B2 16 RGB10
MAIN_DB3 5 A3 B3 15 RGB11
MAIN_DB4 6 A4 B4 14 RGB12
MAIN_DB5 7 A5 B5 13 RGB13
MAIN_DB6 8 A6 B6 12 RGB14
MAIN_DB7 9 A7 B7 11 RGB15
buffers
PALETTECS 1 A->B
COCEH 19 CE
74LS245

RGB8 3 D0 Q0 2 R0
RGB9 4 D1 Q1 5 R1
RGB10 7 D2 Q2 6 R2
RGB11 8 D3 Q3 9 R3
RGB12 13 D4 D20 Q4 12 R4
RGB13 14 D5 Q5 15 G0
RGB14 17 D6 Q6 16 G1
RGB15 18 D7 Q7 19 G2
flip flops
M6 11 Cp
VCC 1 Mr
74LS273

F20
74LS164
High Speed
8-Bit Serial
VBLANK 1 A
HBLANK 2 B AND
M6 8 CLK NOT
WDOGRST 9 MR NOT
Q0 3
Q1 4
Q2 5
Q3 6
Q4 10
Q5 11
Q6 12
Q7 13
In Parallel-Out
Shift Register

C15A 74LS07 1 2
C15B 74LS07 3 4
C15D 74LS07 9 8
C15E 74LS07 11 10
C15F 74LS07 13 12

G4 C17B 74LS07 3 4
G3 C19C 74LS07 5 6
G2 C20F 74LS07 13 12
G1 C19B 74LS07 3 4
G0 C17D 74LS07 9 8

B4 C15E 74LS07 11 10
B3 C15D 74LS07 9 8
B2 C16D 74LS07 9 8
B1 C16E 74LS07 11 10
B0 C17C 74LS07 5 6

R4 C20E 74LS07 11 10
R3 C20B 74LS07 3 4
R2 C20C 74LS07 5 6
R1 C20A 74LS07 1 2
R0 C20D 74LS07 9 8

USED IN THE GAME
LETHAL ENFORCERS

VCC
S2
D4 Pull Up*
D3* D2* D1* D0* BK*
DIGITAL INPUT ANALOG OUTPUT
AO RGB
KONAMI 052535 5-bit Video DAC
VSS

VCC
S1
D4 Pull Up*
D3* D2* D1* D0* BK*
DIGITAL INPUT ANALOG OUTPUT
AO RGB
KONAMI 052535 5-bit Video DAC

VCC
S3
D4 Pull Up*
D3* D2* D1* D0* BK*
DIGITAL INPUT ANALOG OUTPUT
AO RGB
KONAMI 052535 5-bit Video DAC

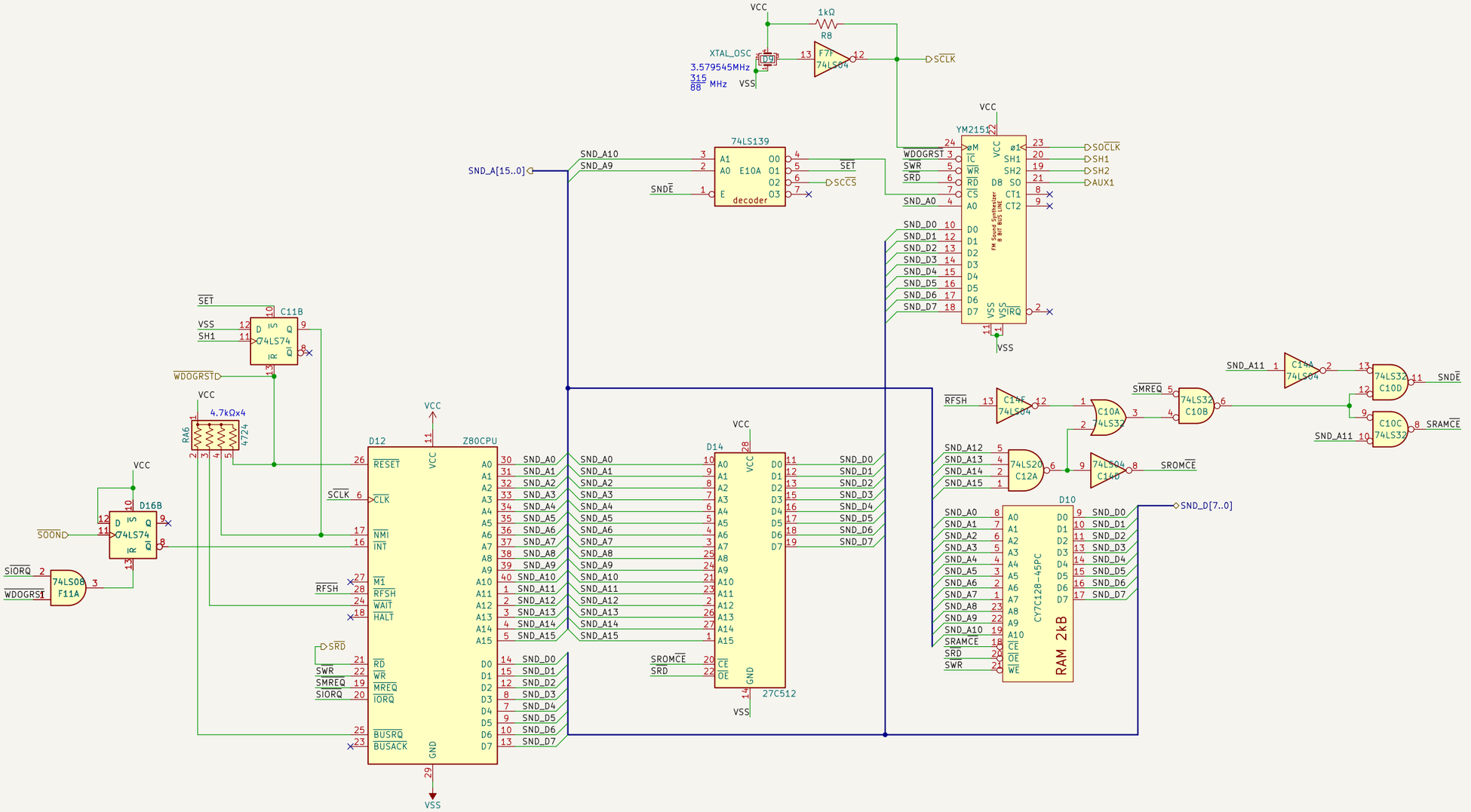
SHADOW 3 C16B 74LS07 4 R21 N.M
MBRIT2 13 C17F 74LS07 12 R16 N.M
GREEN
MBRIT1 13 C19F 74LS07 12 R25 N.M
MBRIT0 11 E19E 74LS07 10 R15 N.M
SHADOW 13 C16F 74LS07 12 R23 N.M
MBRIT2 11 C17E 74LS07 10 R26 N.M
BLUE
MBRIT1 13 74LS00 E16D 11 MBRIT1 11 C19E 74LS07 10 R24 N.M
MBRIT0 5 74LS00 E16B 6 MBRIT0 1 E19A 74LS07 2 R22 118 Ohm
BRITEND 4 74LS00 E16A 3 MBRIT2
MBRIT2 2
MBRIT[2..0]
SHADOW 1 C16A 74LS07 2 R12 N.M
MBRIT2 1 C17A 74LS07 2 R27 N.M
RED
MBRIT1 1 C19A 74LS07 2 R19 N.M
MBRIT0 13 E19F 74LS07 12 R17 N.M

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Sheet: /colmix/rgb/
File: rgb.kicad_sch

Title: Parodius Da!

Size: A3	Date: 2023-05-09	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 7/11



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Sheet: /sound/
File: sound.kicad_sch

Title: Parodius Da!

Size: A3	Date: 2023-05-09	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 8/11

NOT POPULATED

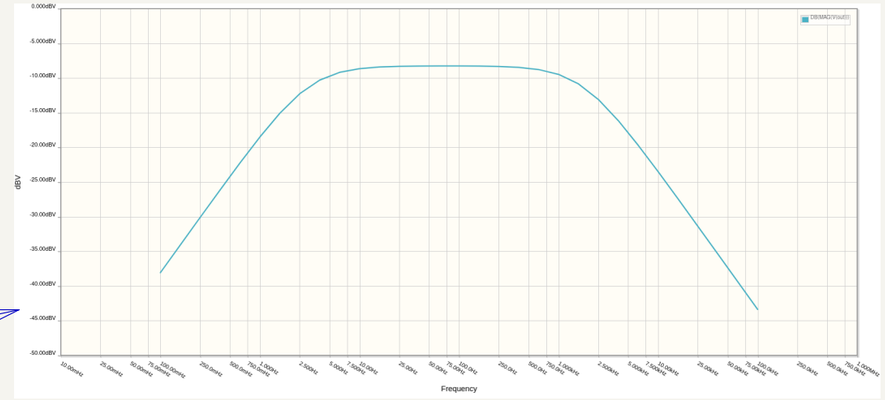
RA[19..0]

Communication between the main CPU
and the audio CPU.

RD[7..0]

Connected to pin 21 of alternate YM2151 footprint.

JP_2_Bridged

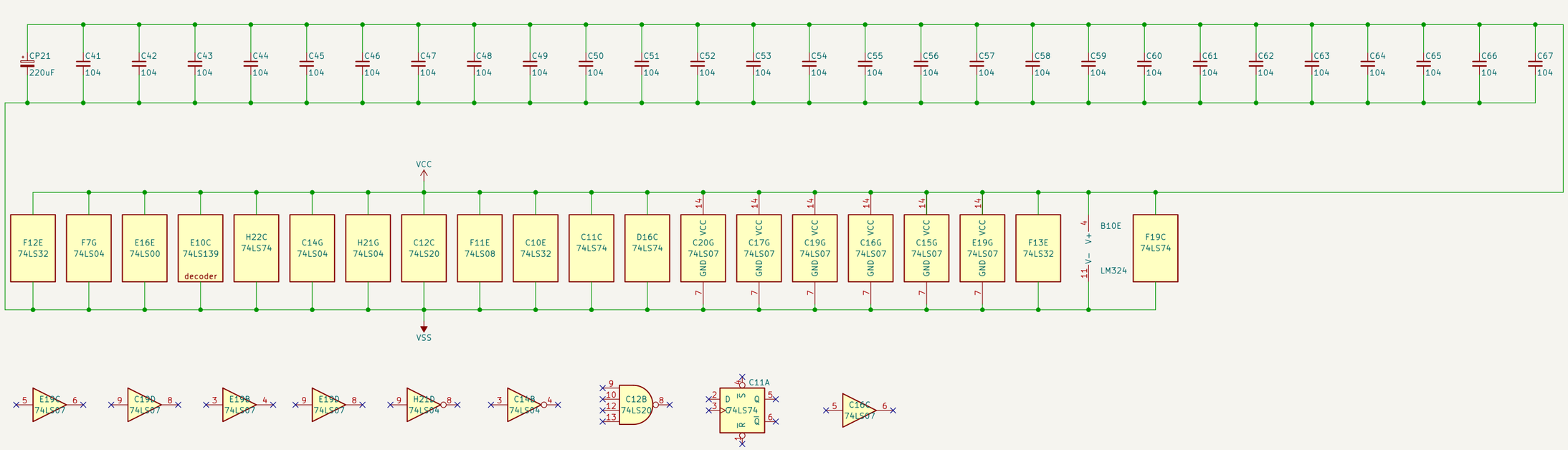


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Sheet: /pcm/
File: pcm.kicad_sch

Title: Parodius Da!

Size: A3	Date: 2023-05-09	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 9/11



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Sheet: /capacitors/
File: capacitors.kicad_sch

Title: Parodius Da!

Size: A3	Date: 2023-05-09	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 10/11

Parodius / Parodius DA!
1990

Parodius! From Myth to Laughter, released in Japan as Parodius Da!
Shinwa kara Owarai e[a] and outside Japan as Parodius
(from "Parody Gradius"), is a shoot 'em up arcade video game and the
second title in the Parodius series produced by Konami.
The European SNES version is also known as Parodius: Non-Sense Fantasy.
The gameplay is stylistically very similar to the Gradius series, but
the graphics and music are intentionally absurd.
As a result of its localised title, From Myth to Laughter is often
mistaken as the original game of the series. The lesser known original game,
Parodius: The Octopus Saves the Earth, was released for the MSX computer in
Japan.

HARDWARE DESCRIPTION

Video Hardware : 052109 051962 053245 053244 053251
CPU : 053248

KNOWN KONAMI TMNT2 BASED GAME LIST
Detana!! Twin Bee / Bells & Whistles / Here Comes Twinbee!!
Golfing Greats
Lightning Fighters / Trigon
Parodius / Parodius DA!
Premier Soccer
Quiz Gakumon no Susume
Rollergames
Sunset Riders
Surprise Attack
Teenage Mutant Ninja Turtles : Turtles in Time

Sequels : Gokujyou Parodius / Fantastic Journey (1994)
Sexy Parodius (1996)

Source: System 16 The Arcade Museum
wikipedia.org/wiki/Parodius_Da!

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JOTEGO

Sheet: /description/
File: description.kicad_sch

Title: Parodius Da!

Size: A3	Date: 2023-05-09	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 11/11